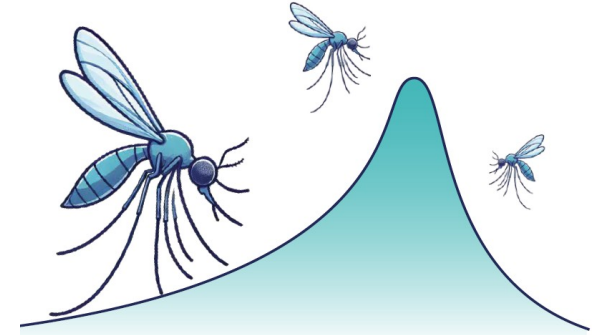


Two decades of *Aedes koreicus* in Europe:
Bridging Knowledge Gaps
in Ecology and Vector Role



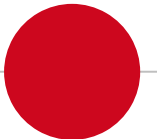
Trento, 10th February 2026

Monitoring and surveillance of *Aedes koreicus* spread

Fabrizio Montarsi

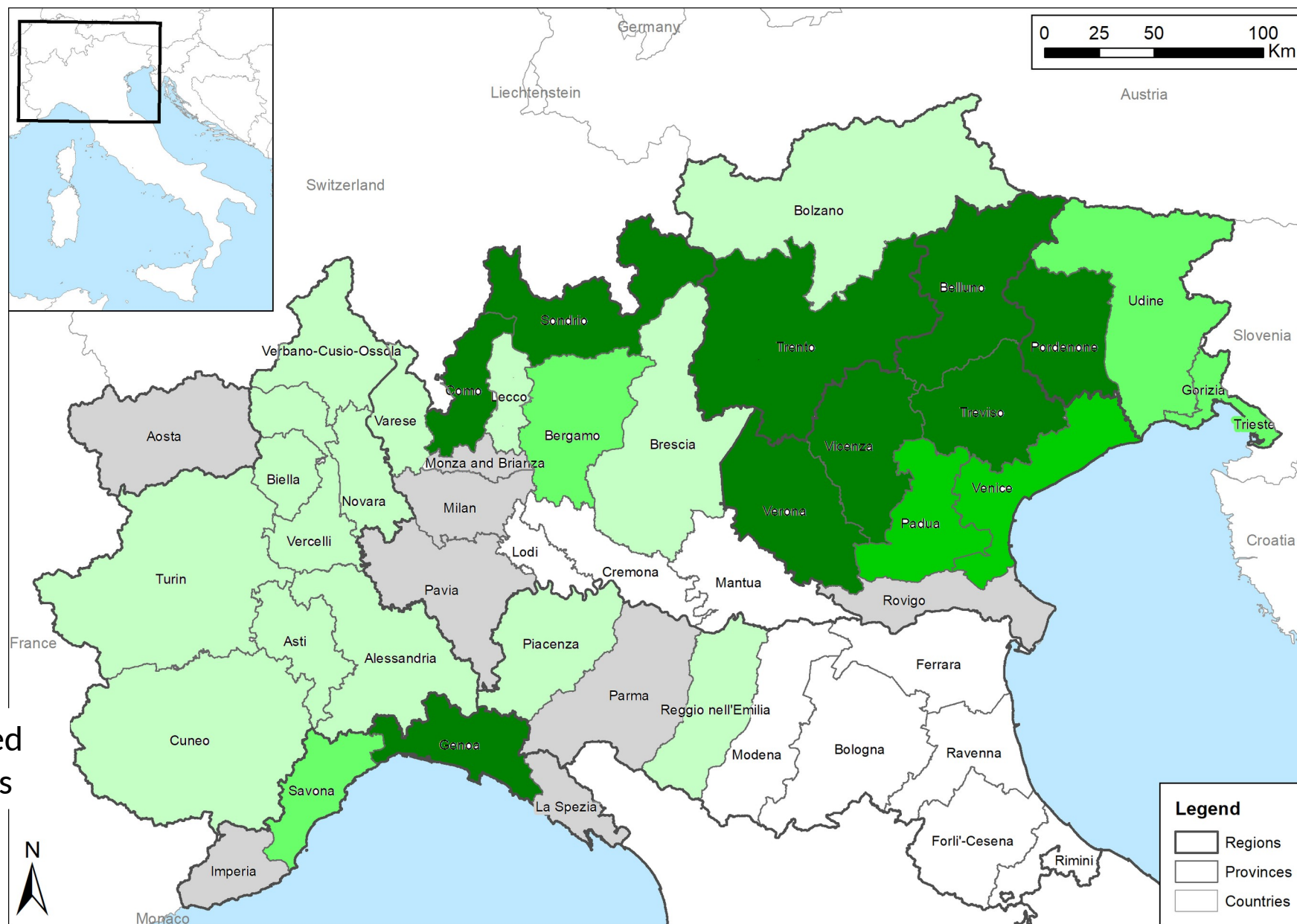
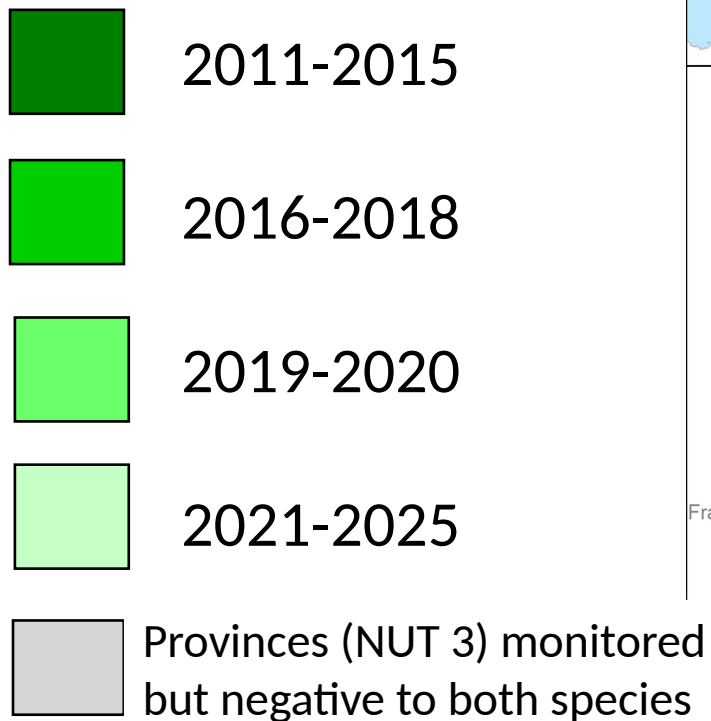
*Laboratory of Medical Entomology and Vector-borne Diseases
Istituto Zooprofilattico Sperimentale delle Venezie, Padua Italy*

Trento, 10 February 2026



Aedes koreicus distribution in Italy

Provinces (NUT 3) positive to
Aedes koreicus
since:



● Guidelines for Aedes invasive mosquitoes monitoring and surveillance

European guidelines for monitoring Invasive mosquitoes

In progress:

*Mosquito Rearing Manual:
A Theoretical and Practical Contribution to Research and Public
Health*

Distribution of Aedes koreicus and Aedes japonicus in Italy

*Standardized Protocols for the Collection, Preservation and
Analysis of invasive mosquitoes (as part of CCM project)*



● Guidelines for Aedes invasive mosquitoes monitoring and surveillance

Italian guidelines for monitoring Invasive mosquitoes

Since 2020

NAZIONAL PLAN FOR PREVENTION, SURVEILLANCE AND RESPONSE TO ARBOVIROSES (PNA) 2020-2025

Drafted by Ministry of Health (Working group)



Adopted by Italian Regions (Department of
Prevention, Food Safety and Veterinary)

Piano Nazionale di prevenzione, sorveglianza e risposta alle Arbovirosi
(PNA) 2020-2025



Novembre 2019



● Guidelines for Aedes invasive mosquitoes monitoring and surveillance

PNA 2020-2025

- (Chapter 6) Surveillance of invasive mosquito species, potential vectors
- Annex 17 – Guidelines for the Identification and Surveillance of Sites at Risk for the Introduction of New Invasive Mosquito Species (excluding *Aedes albopictus*)

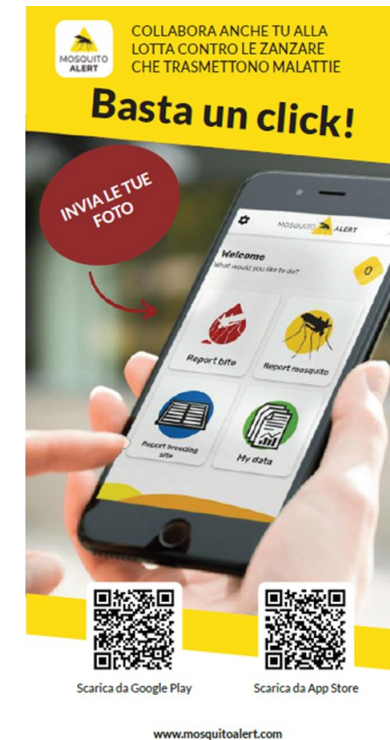
● Aedes invasive mosquitoes monitoring and surveillance in Italy

Other projects that provide data on *Aedes koreicus* occurrence

Entomological surveillance for West Nile and Usutu viruses

Surveillance of Point of Entry

Citizen Science (Mosquito Alert)



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Personnell involved in Aedes Invasive Mosquito (AIM) monitoring

- Local Health Authorities (ASL)
- Private companies helping ASL (Entostudio snc in Noert-Eastern Italy, IPLA in Piedemont region)
- University (Univ. of Milano, Univ. Camerino, ...)
- Other Institution (Fondazione E. Mach, Province of Bolzen,...)
- Istituti Zooprofilattici Sperimentali (IZSVe, IZSER, IZSPLV)
- Istituto Superiore di Sanità
- ISPRA – Italian Institute for Environmental Protection and Research

● Aedes invasive mosquitoes monitoring and surveillance in Italy

Material and Method

Eggs, larvae and adult samplings



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Ovitrap



400 ml

sticky 1L

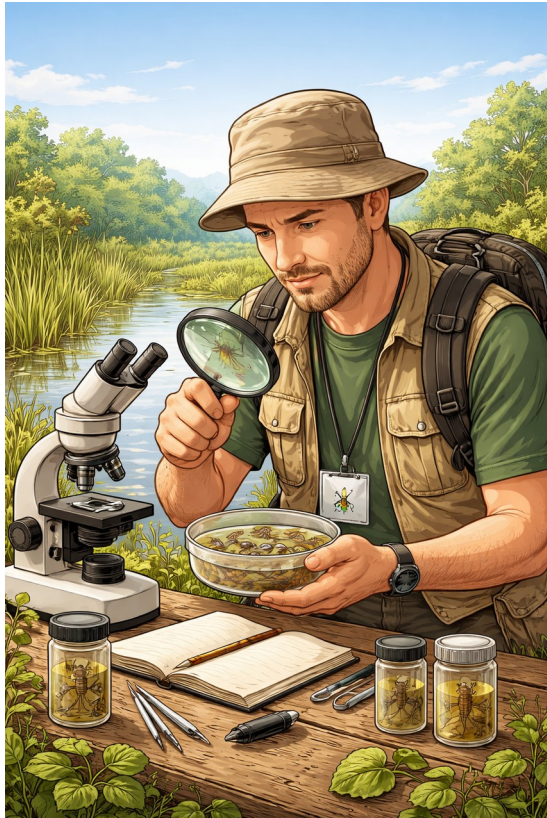
regular 1L

Gravid trap (GAT trap)

● Aedes invasive mosquitoes monitoring and surveillance in Italy

Larvae search

Material



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Larvae search

Material



DATA _____ Operatore _____

CAMPIONAMENTO ZANZARE

COMUNE _____

LOCALITA' _____

ID Sito _____

Descrizione _____

Coordinate N _____ E _____

Altitudine (m asl) _____

Tipo di Luogo	
cimitero	
vivaio	
giardino privato	
parco pubblico	
Gommista	
starda/piazza	
deposito/magazzino	
abitazione/fabbrica	
allevamento	
bosco	
letamaio/discarda	
altro	

Focolaio larvale	
vasi	
sottovasi	
piccoli contenitori artificiali	
grandi contenitori artificiali	
tobini/caditoie	
pneumatici	
ovitrappola	
fontanili	
pozze d'acqua naturali	
scoli di fogne o letamai	
cavità di rocce o alberi	
altro	

Specie	Uova	larve	pupe	adulti
<i>Ae. koreicus</i>				
<i>Ae. albopictus</i>				
<i>Ae. japonicus</i>				
<i>Oc. geniculatus</i>				
<i>Cx. pipiens</i>				
<i>Cx. hortensis</i>				
<i>Anopheles sp.</i>				
<i>Cu. Longiareolata</i>				

Metodo di raccolta	
raccolta acqua manuale	
Dipper	
trappole BG-Sentinel	
CDC-CO2	
Ovitrappole	
aspirazione	
Human landing	
cattura manuale adulti	

Temperatura	
Aria:	Acqua:

Umidità relativa RH%:

● Aedes invasive mosquitoes monitoring and surveillance in Italy

Adult collection



BG-Sentinel trap



CDC-CO₂ trap



Manual aspiration

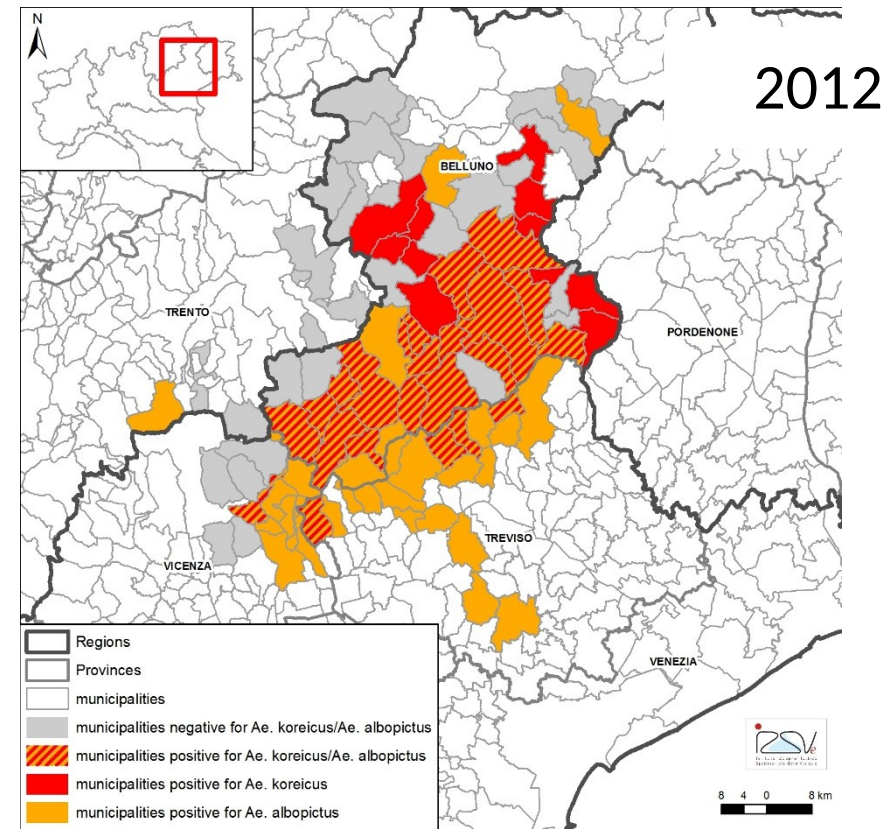
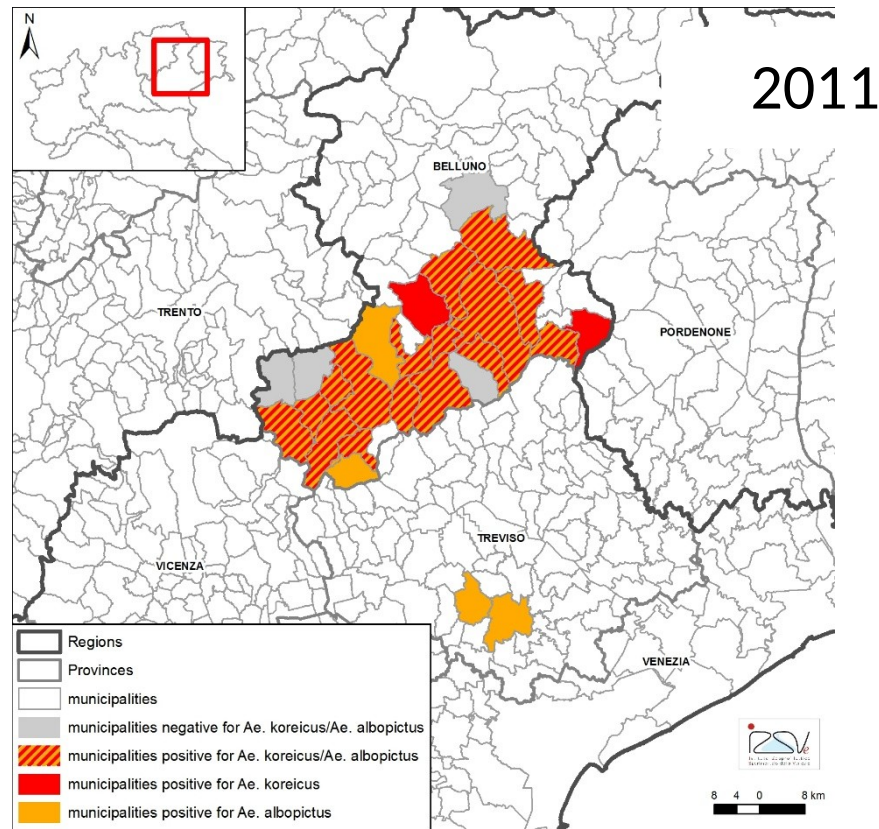


Human landing catch

● Aedes invasive mosquitoes monitoring and surveillance in Italy

Sampling design for *Aedes koreicus*

According to a concentric-area approach



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Breeding site of *Aedes koreicus*



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Breeding site of *Aedes koreicus*



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Breeding site of *Aedes koreicus*



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Environment of *Aedes koreicus*



● Aedes invasive mosquitoes monitoring and surveillance in Italy

Environment of *Aedes koreicus*



● Aedes koreicus monitoring and surveillance in Italy: results

Type of sites

2014

Type of sites	Positive/monitored for <i>Ae. koreicus</i>
Cemeteries	20/101 (19.8%)
Garden centers	13/19 (68.4%)
Private gardens	32/49 (65.3%)
Public parks	2/9 (22.2%)
Tire markets	4/8 (50%)
Streets/squares	28/48 (58.3%)
Depots	4/5 (80%)
Buildings	6/11 (54.5%)
Farms	2/6 (33.3%)
Forests	2/3 (66.7%)
Total	113/259 (43.6%)

2015

Type of sites	Positive/monitored for <i>Ae. koreicus</i>
Cemeteries	21/104 (20.2%)
Garden centers	13/19 (68.4%)
Private gardens	39/56 (69.6%)
Public parks	2/9 (22.2%)
Tire markets	4/8 (50%)
Streets/squares	28/48 (58.3%)
Depots	5/6 (83.3%)
Buildings	6/11 (54.5%)
Farms	2/7 (28.6%)
Forests	2/3 (66.7%)
Total	122/271 (45.0%)



● Aedes koreicus monitoring and surveillance in Italy: results

Breeding sites

2014

Breeding sites	Positive/monitored for <i>Ae. koreicus</i>
Vases	16/93 (17.2%)
Water container	57/139 (41.0%)
Catch basins	18/47 (38.3%)
Puddles	3/6 (50%)
Tires	4/9 (44.4%)
Basin of fountains	6/9 (66.6%)
Dunghill	0/1 (0.0%)
Tree holes	0/1 (0.0%)
Total	104/305 (34.1%)

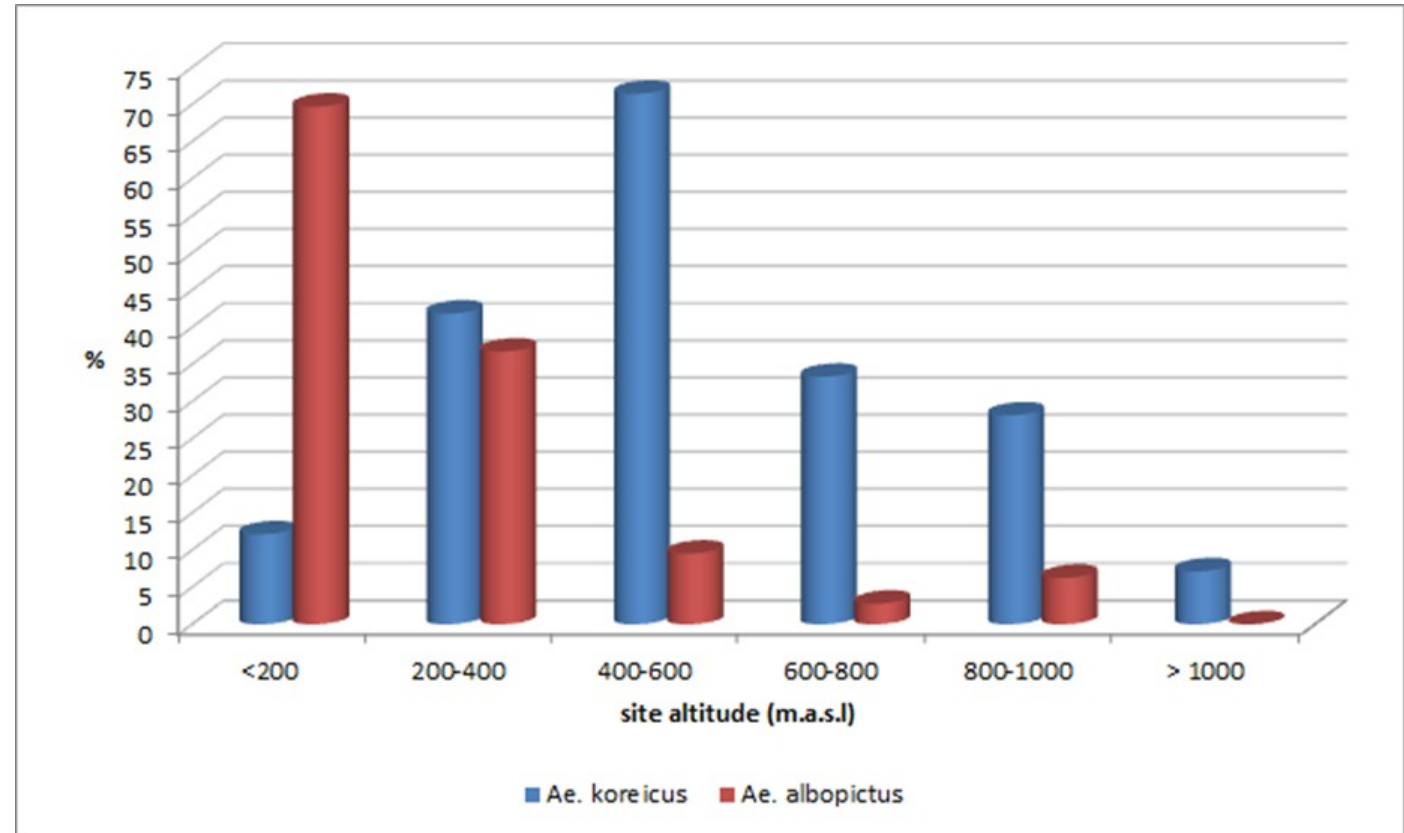
2015

Breeding sites	Positive/monitored for <i>Ae. koreicus</i>
Vases	16/97 (16.5%)
Water container	63/139 (45.3%)
Catch basins	18/48 (37.5%)
Puddles	3/7 (42.8%)
Tires	5/10 (50.0%)
Basin of fountains	7/10 (70.0%)
Dunghill	0/2 (0.0%)
Tree holes	0/1 (0.0%)
Total	112/314 (35.7%)



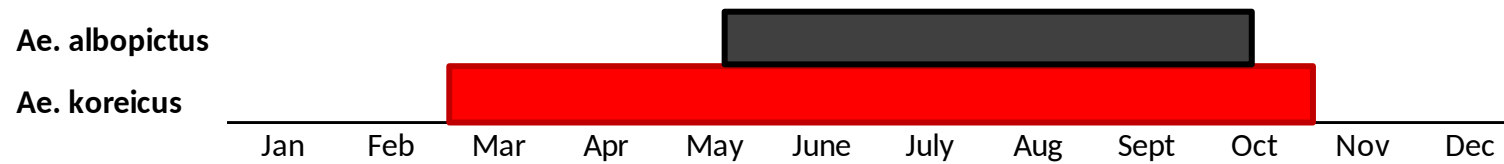
● Aedes koreicus monitoring and surveillance in Italy: results

Altitudinal distribution



● *Aedes koreicus* monitoring and surveillance in Italy: results

Seasonal distribution



Larvae of *Aedes koreicus* found from mid-February to beginning of November



● Conclusions

Larval sampling represents the most effective method for detecting an AIM; however, it is time-consuming, and requires well-trained personnel and available to conduct fieldwork

Aedes koreicus occurs in natural, suburban, and urban environments, but is not typically found in large cities

Breeding sites are mainly artificial containers in rural areas (e.g. vegetable gardens, private gardens, parks, and garden centers), and less frequently in cemeteries



● Conclusions

Larvae of *Aedes koreicus* were found in breeding sites with temperatures ranging between 8 and 27 °C

Larvae of *Aedes koreicus* are frequently found in cohabitation with other mosquito species, including *Aedes albopictus*, *Aedes japonicus*, and *Culex pipiens*

Adult mosquito were found in forested environments and are frequently found inside human dwellings

scientific reports

It feeds on a wide variety of hosts, but is predominantly mammalophilic and exhibits opportunistic feeding behavior



 **First report of the blood-feeding pattern in *Aedes koreicus*, a new invasive species in Europe**

Fabrizio Montarsi¹, Fausta Rosso², Daniele Arnoldi², Silvia Ravagnan¹, Giovanni Marini², Luca Delucchi², Roberto Rosà^{2,3} & Annapaola Rizzoli²

● Conclusions

Aedes koreicus has spread across Italy, initially establishing itself in ecological niches where other mosquito species were poorly represented

The spread advanced from east to west along mountainous and pre-alpine areas, only more recently has it begun to expand southwards and into lowland areas

A separate introduction occurred in Genoa, and (maybe) another one from Switzerland into Lombardy

PLOS NEGLECTED TROPICAL DISEASES

RESEARCH ARTICLE

The genetic trail of the invasive mosquito species *Aedes koreicus* from the east to the west of Northern Italy

Laura Soresinetti¹, Giovanni Naro¹, Irene Arnoldi¹, Andrea Mosca², Katja Adam³, Heung Chul Kim^{4a}, Terry A. Klein^{4a,b}, Francesco Gradoni⁵, Fabrizio Montarsi⁵, Claudio Bandi^{1,6}, Sara Epis^{1,6*}, Paolo Gabrieli^{1,6*}

- These data underline the need for continuous monitoring to track the dynamics of colonization and to support the planning of control strategies against its further spread

Thank you for your attention



<https://www.izsvenezie.com/reference-laboratories/diseases-at-the-animalhuman-interface/>

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